Kabir Singh

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Summary

Computer Science undergraduate with deep interest in reinforcement learning and robotics simulation. Contributed to two open-source ML repositories and completed research internship on autonomous navigation. Seeking roles at the intersection of AI research and applied robotics.

Education

B.Tech in Computer Science (AI Specialization) 2022–2026

IIIT Hyderabad **CGPA: 9.0**

Class XII (2022): 95% | Class X (2020): 96%

Technical Skills

Languages: Python, C++, Rust, SQL

ML/RL: PyTorch, Stable-Baselines3, OpenAI Gym, RLlib, JAX

Robotics: ROS2, Gazebo, MuJoCo, OpenCV

Math: Linear Algebra, Probability, Optimization, Control Theory

Tools: Docker, Git, Weights & Biases, SLURM

Projects

Autonomous Navigation with RL PyTorch, ROS2, Gazebo

- Trained PPO agent for obstacle avoidance in 3D Gazebo environment
- Achieved 88% success rate on unseen maps after curriculum learning
- Integrated trained policy into ROS2 node for real robot deployment

Multi-Agent Stock Trading Simulator RLlib, Python

- Simulated competitive multi-agent environment with 5 RL trading agents
- Compared PPO, SAC, and DDPG algorithms; SAC outperformed others by 12% Sharpe ratio

Open Source: Gym Environment for Traffic Simulation Python, OpenAI Gym

- Created custom Gym environment for traffic signal optimization
- Published on GitHub; 300+ stars and adopted by 2 university research groups

Experience

Research Intern — Autonomous Systems Lab — IIIT Hyderabad Jan 2025 – May 2025

- Implemented model-based RL algorithms (Dreamer v3) for robotic manipulation tasks
- Co-authored workshop paper accepted at ICRA 2025 on sim-to-real transfer

Certifications

- Deep Reinforcement Learning — Udacity Nanodegree
- Robotics: Aerial Robotics — Coursera (UPenn)
- Probabilistic Graphical Models — Coursera (Stanford)